

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	-
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed Smart Lender project aims to transform the traditional loan approval process by using machine learning techniques to improve efficiency, speed, and accuracy.

Project Overview	
Objective	The Smart Lender project aims to modernize the loan approval process by using machine learning to predict whether a loan application will be approved or rejected.
Scope	The scope of the Smart Lender project includes analyzing loan applicant data, building and evaluating machine learning models, and deploying a web application for real-time loan approval prediction.
Problem Statement	
Description	The Smart Lender project addresses delays and inefficiencies in traditional loan approval by using machine learning to predict loan approval based on applicant data.
Impact	The Smart Lender project improves operational efficiency, reduces manual effort and approval delays, and helps financial institutions make faster, more accurate, and consistent lending decisions.
Proposed Solution	
Key Features	The key features of the Smart Lender project include automated loan approval prediction, applicant credibility analysis, data preprocessing, model comparison, and a user-friendly web application for real-time prediction.

	- Automated applicant credibility analysis using historical loan data - Data preprocessing techniques including missing value handling, encoding, scaling, and balancing.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU / Standard system with multi-core processor
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Vs code
Data		
Data	Source, size, format	Kaggle dataset, 614